



CEP-CCIT
FAKULTAS TEKNIK

Implementation of Academic Information System Using Python and MySQL with CIA Triad Security Principle

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CEP CCIT

FAKULTAS TEKNIK UNIVERSITAS INDONESIA

2025

PROJECT ON

***Implementation of Academic Information System Using Python
and MySQL with CIA Triad Security Principle***

Developed by

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Implementation of Academic Information System Using Python and MySQL with CIA Triad Security Principle

Batch Code : 3CS1

Start Date : October 15th, 2025

End Date : October 17th, 2025

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Date of Submission: October 17th, 2025

ACKNOWLEDGEMENT

First and foremost, I would like to express my deepest gratitude to Allah SWT for His endless blessings and guidance that have enabled me to complete this project, titled ***“Implementation of Academic Information System Using Python and MySQL with CIA Triad Security Principle”***

I would like to extend my sincere appreciation to my supervisor, Mr. Ivan Virdaus S.T & Mr. Tri Agus Riyadi, S.Kom., M.T., for their invaluable support, encouragement, and insightful guidance throughout the development of this project. Their expertise and patience played a crucial role in shaping this work.

My heartfelt thanks go to my family and friends for their unwavering support, prayers, and motivation. Their belief in my abilities has always been a source of strength and inspiration.

I am also grateful to my colleagues and everyone who provided constructive feedback, technical assistance, and moral support during the completion of this project.

Finally, my appreciation goes to all those who have directly or indirectly contributed to the success of this work. Thank you for being part of this journey.

Depok, 17 Oktober 2025

BACKGROUND

The development of technology has made information systems an important necessity in facilitating human activities, including in educational institutions. Managing academic data, which includes student data, courses, and Study Plan Cards (KRS), manually is often inefficient and prone to errors. Therefore, the implementation of a technology system that is able to manage data in a structured and efficient manner is a must.

This project aims to develop a simple academic information system built to manage student data and grades. Fundamentally, the system was designed with a primary focus on cybersecurity, referring to the three pillars of information security known as the CIA Triad: Confidentiality, Integrity, and Availability. The concept of Confidentiality is implemented through a multi-user system that separates the roles of Admins and Students, where Students can only access their own personal data and grades, while Integrity is controlled by granting full data CRUD (Create, Read, Update, Delete) rights only to Admins.

The system was developed using the Python programming language and integrated with the MySQL database via XAMPP/phpMyAdmin. The multi-user implementation in this system directly supports security principles, as the clear division of roles will prevent unauthorized data manipulation (Integrity) and ensure that sensitive data is only accessed by authorized parties (Confidentiality). In addition, the use of a stable MySQL database ensures that data is always available (Availability) to authorized users.

SYSTEM ANALYSIS

System Summary :

This project is a simple academic information system built to manage student data and grades. The system was developed using Python programming language and integrated with MySQL database through XAMPP/phpMyAdmin. The system is designed with two types of users, namely administrators and students, who have different access rights, and demonstrates the application of the multi-user concept and the integration of Python with the database, which is the basis for implementing the CIA Triad security principle.

CIA Triad Security Principles and User Functions :

1. Confidentiality

This principle is guaranteed through restricting access to authorized users only:

- **Login Access:** Students login using their NIM; Admins login using a predefined username and password.
- **Personalized Messages:** After a successful student login, the system displays a message: "Login mahasiswa berhasil! Selamat datang, [Student Name]"
- **Limited Access:** Students only have the feature to view their Profile and see the Grades associated with their own NIM.

2. Integrity

This principle is maintained through the full control given to the Admin to maintain the authenticity and accuracy of the data:

- **Full Admin Control:** Admin has full rights to manage (Create, Read, Update, Delete) all data: Students, Courses, and KRS (Grades)
- **Account Automation:** Admin automatically creates login accounts for new students, with NIM serving as the initial username and password.

SYSTEM ANALYSIS

3. Availability

This principle is guaranteed through architectural stability and system functionality:

- **Stable Platform:** The system was developed using a combination of reliable Python and MySQL.
- **Data Access:** Academic data (profile, grades, course list) is available in real-time to authenticated users (Admin and Students).

SOURCE CODE

Main Program:

```
from moduls.auth import login_admin, registrasi_admin, login_mahasiswa
from moduls.mahasiswa import tambah_mahasiswa, lihat_mahasiswa, lihat_info_pribadi
from moduls.nilai import tambah_nilai, lihat_nilai_mahasiswa
from moduls.matakuliah import lihat_matakuliah
from moduls.backup import backup_data

def menu_admin():
    while True:
        print("\n=== Menu Admin ===")
        print("1. Tambah Mahasiswa")
        print("2. Lihat Mahasiswa")
        print("3. Tambah Nilai")
        print("4. Lihat Nilai")
        print("5. Backup Data")
        print("6. Lihat Mata Kuliah")
        print("7. Kembali")
        pilih = input("Pilih: ")

    if pilih == "1":
        tambah_mahasiswa()

    elif pilih == "2":
        lihat_mahasiswa()

    elif pilih == "3":
        tambah_nilai()
```


SOURCE CODE

```
elif pilih == "4":
    nim = input("Masukkan NIM Mahasiswa: ")
    lihat_nilai_mahasiswa(nim)
elif pilih == "5":
    backup_data()
elif pilih == "6":
    lihat_matakuliah()
elif pilih == "7":
    break
else:
    print("Pilihan tidak valid!")

def menu_mahasiswa(nim):
    while True:
        print("\n=== Menu Mahasiswa ===")
        print("1. Lihat Info Pribadi")
        print("2. Lihat Nilai")
        print("3. Lihat Mata Kuliah")
        print("4. Keluar")
        pilih = input("Pilih: ")

        if pilih == "1":
            lihat_info_pribadi(nim)
        elif pilih == "2":
            lihat_nilai_mahasiswa(nim)
        elif pilih == "3":
            lihat_matakuliah()
        elif pilih == "4":
            break
        else:
            print("Pilihan tidak valid!")
```

SOURCE CODE

```
def main():
    while True:
        print("\n=== Sistem Akademik ICA ===")
        print("1. Login Admin")
        print("2. Registrasi Admin")
        print("3. Login Mahasiswa")
        print("4. Keluar")
        pilih = input("Pilih: ")

        if pilih == "1":
            if login_admin():
                menu_admin()
        elif pilih == "2":
            registrasi_admin()
        elif pilih == "3":
            nim = login_mahasiswa()
            if nim:
                menu_mahasiswa(nim)
        elif pilih == "4":
            print("Terima kasih telah menggunakan Sistem Akademik ICA!")
            break
        else:
            print("Pilihan tidak valid!")

if __name__ == "__main__":
    main()
```

OUTPUT PROGRAM

- The figure shows the admin and students tables, both of which use password hashing. If an Admin or Student registers or changes their password, the system automatically stores the hashed result instead of the plain text password. This ensures that user credentials remain secure and unreadable in the database.

				id	username	password
☐	Edit	Copy	Delete	10	ADMIN SA ICA	\$2b\$12\$JvJ/EVHG2qitEL/GzDgqnedt4x246Z7YuPPgEyS5vPB...
☐	Edit	Copy	Delete	11	ADMIN SA ICA2	\$2b\$12\$0gjGAUBon/dFyqllZNDvQ.ANbapHJbLBg0L3iDml4ZK...
☐	Edit	Copy	Delete	12	ADMIN SA ICA3	\$2b\$12\$KLlLaLoCdg7AeF1asd.NUo.1cmRWUWljnvlkh7tjTUAf...
☐	Edit	Copy	Delete	13	Admin SA ICA4	\$2b\$12\$tvf3o0y9MCJ1Rs/fBjjEa.bCqnxOynzXQqxT1Z6.J6o...
☐	Edit	Copy	Delete	15	ADMIN SA ICA5	\$2b\$12\$K1TOdHwE0svAJ1zLVL/a/ep32lwzUjht0wnOBPDVxHQ...
☐	Edit	Copy	Delete	16	ADMIN SA ICA9	\$2b\$12\$e8llGWZppSWyQ/Lzlr2WOdDzccoDu/wXh0/c3nGYF7...
☐	Edit	Copy	Delete	17	AMDIN SA ICA10	\$2b\$12\$.Bp8ld6YvbcOLuOtun6L7OaJ/Y9wBnXfrNNs0B8MARK...

				nim	nama	jurusan	password
☐	Edit	Copy	Delete	1011001	amriganteng	informatika	\$2b\$12\$87seKjBwYVdj3XOtPlgCjONU07V5pPrLullIZGfczEsx...
☐	Edit	Copy	Delete	1011002	nay	Basis data	\$2b\$12\$yyw3k50KvAtYJ/QBgiXp8ury4eJf9hy09DGtPIHkvNR...
☐	Edit	Copy	Delete	1011003	muti	jaringan komputer	\$2b\$12\$SkRn41aCmtLRt22rMGzx0eqfuYFocgCcgGB1iZZ/zbD...
☐	Edit	Copy	Delete	1011009	yamal	Basis data	\$2b\$12\$06kQ4OMeetrneB1aLEkUSe6olbq.iAwRFvhELhUkbZC...
☐	Edit	Copy	Delete	Jaenudin 7			\$2b\$12\$tm9EaybIOP2kHCDuNaStROxJ505KecD7BRqtOmu028R...

- The registration process allows a new Admin to create an account. If the Admin enters a valid username and password, the system confirms with the message “Registrasi admin berhasil!” indicating that the account has been created.

```

=== Sistem Akademik ICA ===
1. Login Admin
2. Registrasi Admin
3. Login Mahasiswa
4. Keluar
Pilih: 2
Masukkan username baru: AMDIN SA ICA10
Masukkan password: amri&nay
Registrasi admin berhasil!

```

OUTPUT PROGRAM

- If the Admin enters the correct username and password, the system displays “Login berhasil! Selamat datang, [Name]” and redirects the Admin to the main menu. Likewise, if a Student enters the correct NIM and password, the system shows “Login mahasiswa berhasil! Selamat datang, [Name]” and displays the main student menu.

```
=== Sistem Akademik ICA ===
1. Login Admin
2. Registrasi Admin
3. Login Mahasiswa
4. Keluar
Pilih: 1
Masukkan username: AMDIN SA ICA10
Masukkan password: amri&nay
Login berhasil! Selamat datang, AMDIN SA ICA10
```

```
=== Sistem Akademik ICA ===
1. Login Admin
2. Registrasi Admin
3. Login Mahasiswa
4. Keluar
Pilih: 3
Masukkan NIM Mahasiswa: 1011001
Masukkan Password: amriganteng
Login mahasiswa berhasil! Selamat datang, amriganteng
```

- If an Admin tries to log in with a username that is not registered, the system displays the message “Username tidak ditemukan.” Similarly, if the user enters an invalid menu option, the system shows “Pilihan tidak valid!” and requests the user to input again.

```
=== Sistem Akademik ICA ===
1. Login Admin
2. Registrasi Admin
3. Login Mahasiswa
4. Keluar
Pilih: 1
Masukkan username: ADMIN SA ICA10
Masukkan password: amri&nay
Username tidak ditemukan.
```

```
=== Sistem Akademik ICA ===
1. Login Admin
2. Registrasi Admin
3. Login Mahasiswa
4. Keluar
Pilih: AMDIN SA ICA10
Pilihan tidak valid!
```

OUTPUT PROGRAM

- If the Admin chooses Add Student, the system requests input such as NIM, name, major, and password. After submission, it confirms with “Mahasiswa berhasil ditambahkan!” meaning the data has been stored in the database.

```
=== Menu Admin ===
1. Tambah Mahasiswa
2. Lihat Mahasiswa
3. Tambah Nilai
4. Lihat Nilai
5. Backup Data
6. Lihat Mata Kuliah
7. Kembali
Pilih: 1
Masukkan NIM: 1011003
Masukkan Nama: muti
Masukkan Jurusan: jaringan komputer
Masukkan Password Mahasiswa: muti
Mahasiswa berhasil ditambahkan!
```

- If the Admin selects View Students, the system retrieves and displays all registered student data, including NIM, name, and major.

```
=== Menu Admin ===
1. Tambah Mahasiswa
2. Lihat Mahasiswa
3. Tambah Nilai
4. Lihat Nilai
5. Backup Data
6. Lihat Mata Kuliah
7. Kembali
Pilih: 2

=== Daftar Mahasiswa ===
NIM: 1011001, Nama: amriganteng, Jurusan: informatika
NIM: 1011002, Nama: nay, Jurusan: Basis data
NIM: 1011003, Nama: muti, Jurusan: jaringan komputer
NIM: 1011009, Nama: yamal, Jurusan: Basis data
NIM: Jaenudin, Nama: 7, Jurusan:
```

OUTPUT PROGRAM

- If the Admin chooses Add Grade, the system asks for the student's NIM, course code, and score. After saving, it shows "Nilai berhasil ditambahkan." confirming that the operation was successful.

```
=== Menu Admin ===
1. Tambah Mahasiswa
2. Lihat Mahasiswa
3. Tambah Nilai
4. Lihat Nilai
5. Backup Data
6. Lihat Mata Kuliah
7. Kembali
Pilih: 3
Masukkan NIM: 1011003
Masukkan Kode Mata Kuliah: IF102
Masukkan Nilai: 78
Nilai berhasil ditambahkan.
```

- If the Admin selects View Grades, the system displays each student's grades by course name and score.

```
=== Menu Admin ===
1. Tambah Mahasiswa
2. Lihat Mahasiswa
3. Tambah Nilai
4. Lihat Nilai
5. Backup Data
6. Lihat Mata Kuliah
7. Kembali
Pilih: 4
Masukkan NIM Mahasiswa: 1011003

=== Nilai Mahasiswa ===
Jaringan Komputer: 78
```

OUTPUT PROGRAM

- If the Admin selects View Courses, the system lists all available courses along with their course codes and credits (SKS).

```
=== Menu Admin ===
1. Tambah Mahasiswa
2. Lihat Mahasiswa
3. Tambah Nilai
4. Lihat Nilai
5. Backup Data
6. Lihat Mata Kuliah
7. Kembali
Pilih: 6

=== Daftar Mata Kuliah ===
IF101 - Pemrograman Dasar (3 SKS)
IF102 - Jaringan Komputer (3 SKS)
IF103 - Keamanan Siber (3 SKS)
IF104 - Basis Data (3 SKS)
IF105 - Kecerdasan Buatan (3 SKS)
IF106 - Sistem Operasi (3 SKS)
=====
```

- If the Admin performs Backup Data, the system automatically saves a backup file (in .sql format) and displays the message “*Backup database berhasil disimpan ke: ...*” confirming the process was successful.

```
=== Menu Admin ===
1. Tambah Mahasiswa
2. Lihat Mahasiswa
3. Tambah Nilai
4. Lihat Nilai
5. Backup Data
6. Lihat Mata Kuliah
7. Kembali
Pilih: 5
Backup database berhasil disimpan ke: SistemAkademikICA/database\akademik_ica.sql
```

OUTPUT PROGRAM

- The activity log table records every action performed by users, both Admins and Students. If any activity occurs—such as login, registration, adding or editing data, or performing a backup—the system automatically stores the timestamp, username, and action details. This feature ensures all activities can be tracked and audited for system security and reliability.

			id	waktu	aktivitas
☐	Edit	Copy	Delete	1	2025-10-17 00:22:42 Percobaan login admin gagal: admin
☐	Edit	Copy	Delete	2	2025-10-17 00:23:55 Admin baru 'ADMIN SA ICA' berhasil diregistrasi.
☐	Edit	Copy	Delete	3	2025-10-17 00:24:13 Admin 'ADMIN SA ICA' berhasil login.
☐	Edit	Copy	Delete	4	2025-10-17 00:24:52 Admin menambahkan mahasiswa baru: 1011001 - amriga...
☐	Edit	Copy	Delete	5	2025-10-17 00:25:22 Admin menambahkan nilai 90.0 untuk NIM 1011001 pad...
☐	Edit	Copy	Delete	6	2025-10-17 00:25:50 Admin menambahkan mahasiswa baru: 1011002 - naysil...
☐	Edit	Copy	Delete	7	2025-10-17 00:26:16 Admin menambahkan nilai 90.0 untuk NIM 1011002 pad...
☐	Edit	Copy	Delete	8	2025-10-17 00:26:29 Backup database dibuat: backup_20251017_002629.sql
☐	Edit	Copy	Delete	9	2025-10-17 00:30:15 Admin baru 'ADMIN SA ICA2' berhasil diregistrasi.
☐	Edit	Copy	Delete	10	2025-10-17 00:30:34 Admin 'ADMIN SA ICA2' berhasil login.
☐	Edit	Copy	Delete	11	2025-10-17 00:31:55 Admin baru 'ADMIN SA ICA' berhasil diregistrasi.
☐	Edit	Copy	Delete	12	2025-10-17 00:32:22 Admin 'ADMIN SA ICA' berhasil login.
☐	Edit	Copy	Delete	13	2025-10-17 00:32:47 Admin baru 'ADMIN SA ICA' berhasil diregistrasi.
☐	Edit	Copy	Delete	14	2025-10-17 00:33:00 Admin 'ADMIN SA ICA' berhasil login.
☐	Edit	Copy	Delete	15	2025-10-17 00:33:50 Admin menambahkan mahasiswa baru: 1011003 - udin
☐	Edit	Copy	Delete	16	2025-10-17 00:34:21 Admin menambahkan nilai 57.0 untuk NIM 1011003 pad...
☐	Edit	Copy	Delete	17	2025-10-17 00:34:34 Backup database dibuat: backup_20251017_003434.sql
☐	Edit	Copy	Delete	18	2025-10-17 00:35:15 Mahasiswa '1011001' berhasil login.
☐	Edit	Copy	Delete	19	2025-10-17 00:37:53 Mahasiswa '1011001' berhasil login.

- If a Student selects View Personal Info, the system displays their personal details such as NIM, name, and major.

```
=== Menu Mahasiswa ===
1. Lihat Info Pribadi
2. Lihat Nilai
3. Lihat Mata Kuliah
4. Keluar
Pilih: 1

=== Info Pribadi Mahasiswa ===
NIM      : 1011001
Nama     : amriganteng
Jurusan  : informatika
```


OUTPUT PROGRAM

- If a Student selects View Grades, the system retrieves and displays their grades entered by the Admin.

```
=== Menu Mahasiswa ===
1. Lihat Info Pribadi
2. Lihat Nilai
3. Lihat Mata Kuliah
4. Keluar
Pilih: 2

=== Nilai Mahasiswa ===
Basis Data: 90
```

- If a Student selects View Courses, the system shows a list of all available courses along with course codes and credits.

```
=== Menu Mahasiswa ===
1. Lihat Info Pribadi
2. Lihat Nilai
3. Lihat Mata Kuliah
4. Keluar
Pilih: 3

=== Daftar Mata Kuliah ===
IF101 - Pemrograman Dasar (3 SKS)
IF102 - Jaringan Komputer (3 SKS)
IF103 - Keamanan Siber (3 SKS)
IF104 - Basis Data (3 SKS)
IF105 - Kecerdasan Buatan (3 SKS)
IF106 - Sistem Operasi (3 SKS)
=====
```

OUTPUT PROGRAM

- This is if the admin wants to delete student data

```
=== Menu Admin ===
1. Tambah Mahasiswa
2. Lihat Mahasiswa
3. Tambah Nilai
4. Lihat Nilai
5. Backup Data
6. Lihat Mata Kuliah
7. Hapus Mahasiswa
8. Kembali
Pilih: 7
Masukkan NIM Mahasiswa yang ingin dihapus: 1011001
Apakah Anda yakin ingin menghapus mahasiswa 'amriganteng' (NIM: 1011001)? (y/n): y
✅ Mahasiswa dan data nilainya berhasil dihapus!
```

- Finally, if the user chooses Exit from the main menu, the system displays the message “Terima kasih telah menggunakan Sistem Akademik ICA!” and safely closes the program.

```
=== Sistem Akademik ICA ===
1. Login Admin
2. Registrasi Admin
3. Login Mahasiswa
4. Keluar
Pilih: 4
Terima kasih telah menggunakan Sistem Akademik ICA!
```

PROJECT DEVELOPER

Develop by : Muhammad Rezky Amri & Naysilla Khasanah

Hardware : Asus Vivobook & Lenovo loq 15

Operating System : Microsoft Window 11

Software : 1. XAMPP/PhpMyAdmin
2. Pycharm
3. Microsoft word
4. Microsoft PowerPoint
5. Github: <https://github.com/muhammadrezkyamri/Implementation-of-Academic-Information-System-Using-Python-and-MySQL-with-CIA-Triad-Security/tree/main>